HW6
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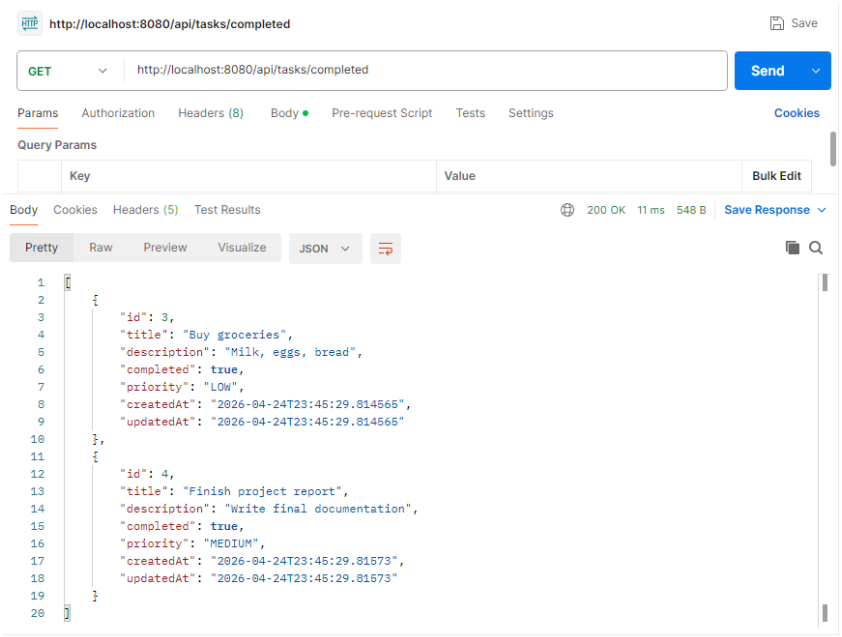
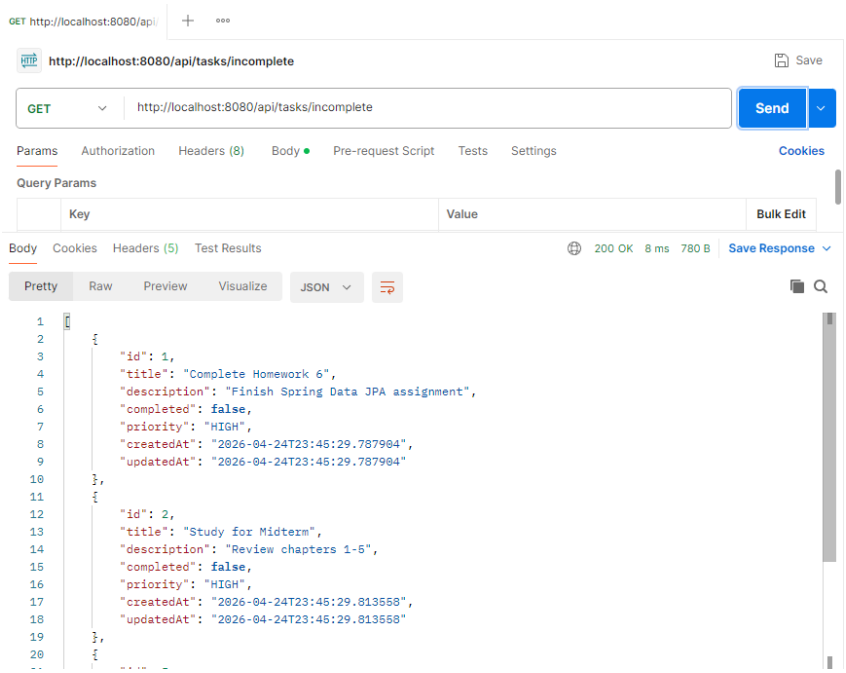
Github: [HW6]

In this assignment, I learned how to use Spring Data JPA to store application data in a database. I learned how to create a JPA entity using annotations like @Entity, @Id, @GeneratedValue, and @Column. I also learned how repository interfaces can automatically generate database queries using method names like findByCompletedTrue() and findByPriority().

Screenshots added below:





GET http://localhost:8080/api

http://localhost:8080/api/tasks/priority/HIGH

Save

GET http://localhost:8080/api/tasks/priority/HIGH

Send

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

Query Params

Key	Value	Bulk Edit

Body Cookies Headers (5) Test Results 200 OK 16 ms 563 B Save Response

Pretty Raw Preview Visualize JSON


```
[
    {
        "id": 1,
        "title": "Complete Homework 6",
        "description": "Finish Spring Data JPA assignment",
        "completed": false,
        "priority": "HIGH",
        "createdAt": "2026-04-24T23:45:29.787904",
        "updatedAt": "2026-04-24T23:45:29.787904"
    },
    {
        "id": 2,
        "title": "Study for Midterm",
        "description": "Review chapters 1-5",
        "completed": false,
        "priority": "HIGH",
        "createdAt": "2026-04-24T23:45:29.813558",
        "updatedAt": "2026-04-24T23:45:29.813558"
    }
]
```


GET http://localhost:8080/api

http://localhost:8080/api/tasks/search?keyword=homework

Save

GET http://localhost:8080/api/tasks/search?keyword=homework

Send

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

☑	keyword	homework
	Key	Value

Body Cookies Headers (5) Test Results 200 OK 11 ms 372 B Save Response

Pretty Raw Preview Visualize JSON


```
[
    {
        "id": 1,
        "title": "Complete Homework 6",
        "description": "Finish Spring Data JPA assignment",
        "completed": false,
        "priority": "HIGH",
        "createdAt": "2026-04-24T23:45:29.787904",
        "updatedAt": "2026-04-24T23:45:29.787904"
    }
]
```

GET http://localhost:8080/api/

http://localhost:8080/api/tasks/paginated?page=0&size=5&sortBy=title

Save

GET http://localhost:8080/api/tasks/paginated?page=0&size=5&sortBy=title Send

Params Authorization Headers (8) Body Pre-request Script Tests Settings Cookies

☑	sortBy	title
	Key	Value

Body Cookies Headers (5) Test Results 200 OK 23 ms 1.44 KB Save Response

Pretty Raw Preview Visualize JSON


```
6            "description": "Milk, eggs, bread",
7            "completed": true,
8            "priority": "LOW",
9            "createdAt": "2026-04-24T23:45:29.814565",
10           "updatedAt": "2026-04-24T23:45:29.814565"
11       },
12       {
13           "id": 1,
14           "title": "Complete Homework 6",
15           "description": "Finish Spring Data JPA assignment",
16           "completed": false,
17           "priority": "HIGH",
18           "createdAt": "2026-04-24T23:45:29.787904",
19           "updatedAt": "2026-04-24T23:45:29.787904"
20       },
21       {
22           "id": 4,
23           "title": "Finish project report",
24           "description": "Write final documentation",
25           "completed": true,
```


localhost:8080/h2-console/login.do?jsessionid=3dcb6793d4c90eb8dd5c95fb08056bbb

Gmail YouTube Sir John Wilson Sch... Weebly Wordpress ChemTalk-Trello ChemTalk-Wordpress GSU Health PAWS Mess

Auto commit Max rows: 1000 Auto complete Off Auto select On

jdbc:h2:mem:taskboarddb
- TASKS
- INFORMATION_SCHEMA
- Users

H2 2.2.224 (2023-09-17)

Run Run Selected Auto complete Clear SQL statement:

```
SHOW TABLES;
```

SHOW TABLES;

TABLE_NAME	TABLE_SCHEMA
TASKS	PUBLIC

(1 row, 0 ms)

localhost:8080/h2-console/login.do?jsessionid=3dcb6793d4c90eb8dd5c95fb08056bbb

Gmail YouTube Sir John Wilson Sch... Weebly Wordpress ChemTalk-Trello ChemTalk-Wordpress GSU Health PAWS Messenger Ig

Auto commit | Max rows: 1000 | Auto complete Off | Auto select On

jdbc:h2:mem:taskboarddb
- TASKS
- INFORMATION_SCHEMA
- Users

H2 2.2.224 (2023-09-17)

Run | Run Selected | Auto complete | Clear | SQL statement:

```
SELECT * FROM tasks;
```

SELECT * FROM tasks;

ID	COMPLETED	CREATED_AT	DESCRIPTION	PRIORITY	TITLE	UPDATED_AT
1	FALSE	2026-04-24 23:45:29.787904	Finish Spring Data JPA assignment	HIGH	Complete Homework 6	2026-04-24 23:45:29.787904
2	FALSE	2026-04-24 23:45:29.813558	Review chapters 1-5	HIGH	Study for Midterm	2026-04-24 23:45:29.813558
3	TRUE	2026-04-24 23:45:29.814565	Milk, eggs, bread	LOW	Buy groceries	2026-04-24 23:45:29.814565
4	TRUE	2026-04-24 23:45:29.81573	Write final documentation	MEDIUM	Finish project report	2026-04-24 23:45:29.81573
5	FALSE	2026-04-24 23:45:29.81676	Review controllers, services, repositories	MEDIUM	Practice Spring Boot	2026-04-24 23:45:29.81676

(5 rows, 0 ms)

Edit

```
        lower(t1_0.title) like lower(('%'||?||'%')) escape ''
        or lower(t1_0.description) like lower(('%'||?||'%')) escape ''
Hibernate:
    select
        t1_0.id,
        t1_0.completed,
        t1_0.created_at,
        t1_0.description,
        t1_0.priority,
        t1_0.title,
        t1_0.updated_at
    from
        tasks t1_0
    order by
        t1_0.title
    offset
        ? rows
    fetch
        first ? rows only
Hibernate:
    select
```